

Exame de Seleção para Capitão Amador

Abril / 2019

RELAÇÃO DE ANEXOS

- A) Cópia do marcador de páginas, face correspondente à correção de altura de 10° a 90° – Sol, Estrelas e Planetas, do Almanaque Náutico 2019.
- B) Cópia da página 89 do Almanaque Náutico 2019.
- C) Cópia da página XVI – Acréscimos e Correções do Almanaque Náutico 2019.
- D) Cópia da página I – Conversão de Arco em Tempo, do Almanaque Náutico 2019.
- E) Uma Rosa de Manobra.
- F) Folha em branco para rascunho.

EXAME DE SELEÇÃO PARA CATEGORIA DE CAPITÃO-AMADOR

Exame: I / 2019

Anexos

A2 CORREÇÃO DE ALTURA DE 10° - 90° - SOL, ESTRELAS E PLANETAS

Out — Mar			SOL			Abr — Set			ESTRELAS E PLANETAS			DEPRESSÃO		
a	Limbo		a	Limbo		a	Limbo		a	Corr.		Elev do	Corr.	Elev do
ap	Inf	Sup	ap	Inf	Sup	ap	Inf	Sup	ap	ap	adicional	Olho	Olho	Olho
9 33	+ 10.8	- 21.5	9 39	+ 10.6	- 21.2	9 55	- 5.3		9 55	- 5.3		m		Pés
9 45	+ 10.9	- 21.4	9 50	+ 10.7	- 21.1	10 07	- 5.2		10 07	- 5.2		2.4	- 2.8	8.0
9 56	+ 11.0	- 21.3	10 02	+ 10.8	- 21.0	10 20	- 5.1		10 20	- 5.1		2.6	- 2.9	8.6
10 08	+ 11.1	- 21.2	10 14	+ 10.9	- 20.9	10 32	- 5.0		10 32	- 5.0		2.8	- 3.0	9.2
10 20	+ 11.2	- 21.1	10 27	+ 11.0	- 20.8	10 46	- 4.9		10 46	- 4.9		3.0	- 3.1	9.8
10 33	+ 11.3	- 21.0	10 40	+ 11.1	- 20.7	10 59	- 4.8		10 59	- 4.8		3.2	- 3.2	10.5
10 46	+ 11.4	- 20.9	10 53	+ 11.2	- 20.6	11 14	- 4.7		11 14	- 4.7		3.4	- 3.3	11.2
11 00	+ 11.5	- 20.8	11 07	+ 11.3	- 20.5	11 29	- 4.6		11 29	- 4.6		3.6	- 3.4	11.9
11 15	+ 11.6	- 20.7	11 22	+ 11.4	- 20.4	11 44	- 4.5		11 44	- 4.5		3.8	- 3.5	12.6
11 30	+ 11.7	- 20.6	11 37	+ 11.5	- 20.3	12 00	- 4.4		12 00	- 4.4		4.0	- 3.6	13.3
11 45	+ 11.8	- 20.5	11 53	+ 11.6	- 20.2	12 17	- 4.3		12 17	- 4.3		4.3	- 3.7	14.1
12 01	+ 11.9	- 20.4	12 10	+ 11.7	- 20.1	12 35	- 4.2		12 35	- 4.2		4.5	- 3.8	14.9
12 18	+ 12.0	- 20.3	12 27	+ 11.8	- 20.0	12 53	- 4.1		12 53	- 4.1		4.7	- 3.9	15.7
12 36	+ 12.1	- 20.2	12 45	+ 11.9	- 19.9	13 12	- 4.0		13 12	- 4.0		5.0	- 4.0	16.5
12 54	+ 12.2	- 20.1	13 04	+ 12.0	- 19.8	13 32	- 3.9		13 32	- 3.9		5.2	- 4.1	17.4
13 14	+ 12.3	- 20.0	13 24	+ 12.1	- 19.7	13 53	- 3.8		13 53	- 3.8		5.5	- 4.2	18.3
13 34	+ 12.4	- 19.9	13 44	+ 12.2	- 19.6	14 16	- 3.7		14 16	- 3.7		5.8	- 4.3	19.1
13 55	+ 12.5	- 19.8	14 06	+ 12.3	- 19.5	14 39	- 3.6		14 39	- 3.6		6.1	- 4.4	20.1
14 17	+ 12.6	- 19.7	14 29	+ 12.4	- 19.4	15 03	- 3.5		15 03	- 3.5		6.3	- 4.5	21.0
14 41	+ 12.7	- 19.6	14 53	+ 12.5	- 19.3	15 29	- 3.4		15 29	- 3.4		6.6	- 4.6	22.0
15 05	+ 12.8	- 19.5	15 18	+ 12.6	- 19.2	15 56	- 3.3		15 56	- 3.3		6.9	- 4.7	22.9
15 31	+ 12.9	- 19.4	15 45	+ 12.7	- 19.1	16 25	- 3.2		16 25	- 3.2		7.2	- 4.8	23.9
15 59	+ 13.0	- 19.3	16 13	+ 12.8	- 19.0	16 55	- 3.1		16 55	- 3.1		7.5	- 4.9	24.9
16 27	+ 13.1	- 19.2	16 43	+ 12.9	- 18.9	17 27	- 3.0		17 27	- 3.0		7.9	- 5.0	26.0
16 58	+ 13.2	- 19.1	17 14	+ 13.0	- 18.8	18 01	- 2.9		18 01	- 2.9		8.2	- 5.1	27.1
17 30	+ 13.3	- 19.0	17 47	+ 13.1	- 18.7	18 37	- 2.8		18 37	- 2.8		8.5	- 5.2	28.1
18 05	+ 13.4	- 18.9	18 23	+ 13.2	- 18.6	19 16	- 2.7		19 16	- 2.7		8.8	- 5.3	29.2
18 41	+ 13.5	- 18.8	19 00	+ 13.3	- 18.5	19 56	- 2.6		19 56	- 2.6		9.2	- 5.4	30.4
19 20	+ 13.6	- 18.7	19 41	+ 13.4	- 18.4	20 40	- 2.5		20 40	- 2.5		9.5	- 5.5	31.5
20 02	+ 13.7	- 18.6	20 24	+ 13.5	- 18.3	21 27	- 2.4		21 27	- 2.4		9.9	- 5.6	32.7
20 46	+ 13.8	- 18.5	21 10	+ 13.6	- 18.2	22 17	- 2.3		22 17	- 2.3		10.3	- 5.7	33.9
21 34	+ 13.9	- 18.4	21 59	+ 13.7	- 18.1	23 11	- 2.2		23 11	- 2.2		10.6	- 5.8	35.1
22 25	+ 14.0	- 18.3	22 52	+ 13.8	- 18.0	24 09	- 2.1		24 09	- 2.1		11.0	- 5.9	36.3
23 20	+ 14.1	- 18.2	23 49	+ 13.9	- 17.9	25 12	- 2.0		25 12	- 2.0		11.4	- 6.0	37.6
24 20	+ 14.2	- 18.1	24 51	+ 14.0	- 17.8	26 20	- 1.9		26 20	- 1.9		11.8	- 6.1	38.9
25 24	+ 14.3	- 18.0	25 58	+ 14.1	- 17.7	27 34	- 1.8		27 34	- 1.8		12.2	- 6.2	40.1
26 34	+ 14.4	- 17.9	27 11	+ 14.2	- 17.6	28 54	- 1.7		28 54	- 1.7		12.6	- 6.3	41.5
27 50	+ 14.5	- 17.8	28 31	+ 14.3	- 17.5	30 22	- 1.6		30 22	- 1.6		13.0	- 6.4	42.8
29 13	+ 14.6	- 17.7	29 58	+ 14.4	- 17.4	31 58	- 1.5		31 58	- 1.5		13.4	- 6.5	44.2
30 44	+ 14.7	- 17.6	31 33	+ 14.5	- 17.3	33 43	- 1.4		33 43	- 1.4		13.8	- 6.6	45.5
32 24	+ 14.8	- 17.5	33 18	+ 14.6	- 17.2	35 38	- 1.3		35 38	- 1.3		14.2	- 6.7	46.9
34 15	+ 14.9	- 17.4	35 15	+ 14.7	- 17.1	37 45	- 1.2		37 45	- 1.2		14.7	- 6.8	48.4
36 17	+ 15.0	- 17.3	37 24	+ 14.8	- 17.0	40 06	- 1.1		40 06	- 1.1		15.1	- 6.9	49.8
38 34	+ 15.1	- 17.2	39 48	+ 14.9	- 16.9	42 42	- 1.0		42 42	- 1.0		15.5	- 7.0	51.3
41 06	+ 15.2	- 17.1	42 28	+ 15.0	- 16.8	45 34	- 0.9		45 34	- 0.9		16.0	- 7.1	52.8
43 56	+ 15.3	- 17.0	45 29	+ 15.1	- 16.7	48 45	- 0.8		48 45	- 0.8		16.5	- 7.2	54.3
47 07	+ 15.4	- 16.9	48 52	+ 15.2	- 16.6	52 16	- 0.7		52 16	- 0.7		16.9	- 7.3	55.8
50 43	+ 15.5	- 16.8	52 41	+ 15.3	- 16.5	56 09	- 0.6		56 09	- 0.6		17.4	- 7.4	57.4
54 46	+ 15.6	- 16.7	56 59	+ 15.4	- 16.4	60 26	- 0.5		60 26	- 0.5		17.9	- 7.5	58.9
59 21	+ 15.7	- 16.6	61 50	+ 15.5	- 16.3	65 06	- 0.4		65 06	- 0.4		18.4	- 7.6	60.5
64 28	+ 15.8	- 16.5	67 15	+ 15.6	- 16.2	70 09	- 0.3		70 09	- 0.3		18.8	- 7.7	62.1
70 10	+ 15.9	- 16.4	73 14	+ 15.7	- 16.1	75 32	- 0.2		75 32	- 0.2		19.3	- 7.8	63.8
76 24	+ 16.0	- 16.3	79 42	+ 15.8	- 16.0	81 12	- 0.1		81 12	- 0.1		19.8	- 7.9	65.4
83 05	+ 16.1	- 16.2	86 31	+ 15.9	- 15.9	87 03	0.0		87 03	0.0		20.4	- 8.0	67.1
90 00			90 00			90 00			90 00			20.9	- 8.1	68.8
												21.4	- 8.1	70.5

a ap = Altura dada pelo sextante corrigida do erro instrumental e da depressão

TU	SOL		LUA						Lat.	CREP		SOL	LUA - Nascer				
	AHG		AHG		Dec		d			Naut.	Civil	Nascer	25	26	27	28	
	h	m	h	m	h	m	h	m				h m	h m	h m	h m	h m	
QUINTA FEIRA	00	180 28.4	N13 01.4	288 34.4	10.1	S22 07.2	0.4	55.4	N 72	////	////	02 33	██	██	██	05 49	
	01	195 28.5	02.2	303 03.5	10.2	22 06.8	0.4	55.4	N 70	////	00 57	03 01	██	██	05 36	05 01	
	02	210 28.6	03.0	317 32.7	10.2	22 06.4	0.6	55.3	68	////	01 54	03 22	██	04 44	04 36	04 30	
	03	225 28.7	03.9	332 01.9	10.2	22 05.8	0.7	55.3	66	////	02 26	03 38	03 18	03 48	04 01	04 07	
	04	240 28.8	04.7	346 31.1	10.3	22 05.1	0.8	55.3	64	00 50	02 49	03 52	02 38	03 15	03 36	03 49	
	05	255 29.0	05.5	1 00.4	10.3	22 04.3	0.9	55.3	62	01 40	03 08	04 03	02 11	02 50	03 16	03 34	
	06	270 29.1	N13 06.3	15 29.7	10.4	S22 03.4	1.1	55.2	60	02 10	03 22	04 13	01 50	02 31	03 00	03 21	
	07	285 29.2	07.1	29 59.1	10.3	22 02.3	1.1	55.2	N 58	02 31	03 35	04 21	01 32	02 15	02 47	03 10	
	08	300 29.3	07.9	44 28.4	10.5	22 01.2	1.2	55.2	56	02 49	03 46	04 29	01 18	02 01	02 35	03 01	
	09	315 29.4	08.8	58 57.9	10.5	22 00.0	1.4	55.2	54	03 03	03 55	04 35	01 05	01 49	02 24	02 52	
	10	330 29.5	09.6	73 27.4	10.5	21 58.6	1.5	55.1	52	03 15	04 03	04 41	00 54	01 39	02 15	02 45	
	11	345 29.6	10.4	87 56.9	10.5	21 57.1	1.5	55.1	50	03 26	04 11	04 46	00 44	01 29	02 07	02 38	
	12	0 29.7	N13 11.2	102 26.4	10.6	S21 55.6	1.7	55.1	45	03 47	04 26	04 58	00 23	01 10	01 49	02 23	
	13	15 29.8	12.0	116 56.0	10.7	21 53.9	1.8	55.1	N 40	04 03	04 38	05 07	00 06	00 54	01 35	02 11	
	14	30 29.9	12.8	131 25.7	10.6	21 52.1	1.9	55.1	35	04 17	04 49	05 15	24 40	00 40	01 23	02 01	
	15	45 30.0	13.6	145 55.3	10.8	21 50.2	2.0	55.0	30	04 27	04 57	05 22	24 28	00 28	01 12	01 51	
	16	60 30.1	14.5	160 25.1	10.7	21 48.2	2.1	55.0	20	04 45	05 11	05 34	24 08	00 08	00 54	01 36	
	17	75 30.3	15.3	174 54.8	10.8	21 46.1	2.2	55.0	N 10	04 58	05 23	05 45	23 50	24 37	00 37	01 22	
	18	90 30.4	N13 16.1	189 24.6	10.9	S21 43.9	2.3	55.0	0	05 09	05 33	05 54	23 34	24 22	00 22	01 09	
	19	105 30.5	16.9	203 54.5	10.9	21 41.6	2.4	54.9	S 10	05 18	05 43	06 04	23 17	24 07	00 07	00 56	
	20	120 30.6	17.7	218 24.4	10.9	21 39.2	2.6	54.9	20	05 26	05 52	06 14	23 00	23 51	24 42	00 42	
	21	135 30.7	18.5	232 54.3	11.0	21 36.6	2.6	54.9	30	05 33	06 01	06 26	22 40	23 33	24 26	00 26	
	22	150 30.8	19.3	247 24.3	11.0	21 34.0	2.7	54.9	35	05 36	06 06	06 32	22 28	23 22	24 17	00 17	
	23	165 30.9	20.1	261 54.3	11.1	21 31.3	2.9	54.9	40	05 40	06 12	06 40	22 14	23 09	24 06	00 06	
SEXTA FEIRA	00	180 31.0	N13 20.9	276 24.4	11.1	S21 28.4	2.9	54.8	45	05 43	06 18	06 48	21 58	22 55	23 54	24 54	
	01	195 31.1	21.8	290 54.5	11.2	21 25.5	3.0	54.8	S 50	05 47	06 25	06 59	21 38	22 37	23 39	24 42	
	02	210 31.2	22.6	305 24.7	11.2	21 22.5	3.1	54.8	52	05 48	06 28	07 03	21 28	22 28	23 31	24 36	
	03	225 31.3	23.4	319 54.9	11.2	21 19.4	3.3	54.8	54	05 50	06 31	07 09	21 18	22 19	23 23	24 30	
	04	240 31.4	24.2	334 25.1	11.3	21 16.1	3.3	54.8	56	05 51	06 35	07 14	21 06	22 08	23 14	24 23	
	05	255 31.5	25.0	348 55.4	11.4	21 12.8	3.4	54.8	58	05 53	06 39	07 21	20 52	21 55	23 04	24 16	
	06	270 31.6	N13 25.8	3 25.8	11.4	S21 09.4	3.6	54.7	S 60	05 55	06 44	07 28	20 35	21 41	22 53	24 07	
	07	285 31.7	26.6	17 56.2	11.4	21 05.8	3.6	54.7	Lat. SOL CREP LUA - Pôr								
	08	300 31.8	27.4	32 26.6	11.5	21 02.2	3.7	54.7	Pôr		Civil Naut.		25	26	27	28	
	09	315 31.9	28.2	46 57.1	11.6	20 58.5	3.9	54.7	h m		h m		h m	h m	h m	h m	
	10	330 32.0	29.0	61 27.7	11.5	20 54.6	3.9	54.7	N 72	21 28	////	////	██	██	██	09 05	
	11	345 32.1	29.8	75 58.2	11.7	20 50.7	4.0	54.6	N 70	20 59	23 15	////	██	██	07 42	09 52	
	12	0 32.2	N13 30.6	90 28.9	11.7	S20 46.7	4.1	54.6	68	20 37	22 08	////	██	06 53	08 41	10 22	
	13	15 32.3	31.4	104 59.6	11.7	20 42.6	4.2	54.6	66	20 20	21 34	////	06 35	07 49	09 15	10 44	
	14	30 32.4	32.2	119 30.3	11.8	20 38.4	4.3	54.6	64	20 06	21 10	23 20	07 14	08 22	09 39	11 01	
	15	45 32.5	33.1	134 01.1	11.8	20 34.1	4.4	54.6	62	19 54	20 51	22 21	07 42	08 46	09 59	11 16	
	16	60 32.6	33.9	148 31.9	11.9	20 29.7	4.5	54.6	60	19 45	20 35	21 50	08 03	09 05	10 14	11 28	
	17	75 32.8	34.7	163 02.8	11.9	20 25.2	4.5	54.6	N 58	19 36	20 23	21 27	08 20	09 21	10 28	11 38	
	18	90 32.9	N13 35.5	177 33.7	11.9	S20 20.7	4.7	54.5	56	19 29	20 12	21 09	08 35	09 34	10 39	11 47	
	19	105 33.0	36.3	192 04.6	12.1	20 16.0	4.8	54.5	54	19 22	20 02	20 55	08 48	09 46	10 49	11 55	
	20	120 33.1	37.1	206 35.7	12.0	20 11.2	4.8	54.5	52	19 16	19 54	20 42	08 59	09 56	10 58	12 02	
	21	135 33.2	37.9	221 06.7	12.1	20 06.4	5.0	54.5	50	19 10	19 46	20 32	09 09	10 05	11 06	12 09	
	22	150 33.3	38.7	235 37.8	12.2	20 01.4	5.0	54.5	45	18 59	19 31	20 10	09 29	10 25	11 23	12 23	
	23	165 33.4	39.5	250 09.0	12.2	19 56.4	5.1	54.5	N 40	18 49	19 18	19 53	09 46	10 40	11 37	12 34	
SÁBADO	00	180 33.5	N13 40.3	264 40.2	12.3	S19 51.3	5.2	54.5	35	18 41	19 08	19 40	10 00	10 54	11 48	12 44	
	01	195 33.6	41.1	279 11.5	12.3	19 46.1	5.3	54.4	30	18 34	18 59	19 29	10 12	11 05	11 58	12 52	
	02	210 33.7	41.9	293 42.8	12.3	19 40.8	5.4	54.4	20	18 22	18 45	19 11	10 33	11 25	12 16	13 06	
	03	225 33.8	42.7	308 14.1	12.4	19 35.4	5.5	54.4	N 10	18 11	18 33	18 58	10 51	11 42	12 31	13 19	
	04	240 33.9	43.5	322 45.5	12.5	19 29.9	5.5	54.4	0	18 01	18 23	18 47	11 08	11 58	12 45	13 31	
	05	255 34.0	44.3	337 17.0	12.5	19 24.4	5.7	54.4	S 10	17 52	18 13	18 38	11 25	12 13	12 59	13 42	
	06	270 34.1	N13 45.1	351 48.5	12.5	S19 18.7	5.7	54.4	20	17 41	18 04	18 30	11 43	12 30	13 14	13 55	
	07	285 34.1	45.9	6 20.0	12.6	19 13.0	5.8	54.4	30	17 30	17 54	18 22	12 04	12 50	13 31	14 09	
	08	300 34.2	46.7	20 51.6	12.6	19 07.2	5.9	54.4	35	17 23	17 49	18 19	12 16	13 01	13 41	14 17	
	09	315 34.3	47.5	35 23.2	12.7	19 01.3	6.0	54.4	40	17 16	17 43	18 15	12 30	13 14	13 52	14 26	
	10	330 34.4	48.3	49 54.9	12.8	18 55.3	6.0	54.4	45	17 07	17 37	18 12	12 46	13 29	14 06	14 37	
	11	345 34.5	49.1	64 26.7	12.8	18 49.3	6.2	54.3	S 50	16 56	17 30	18 08	13 06	13 48	14 22	14 50	
	12	0 34.6	N13 49.9	78 58.5	12.8	S18 43.1	6.2	54.3	52	16 52	17 27	18 07	13 16	13 56	14 29	14 56	
	13	15 34.7	50.7	93 30.3	12.9	18 36.9	6.3	54.3	54	16 46	17 23	18 05	13 27	14 06	14 38	15 03	
	14	30 34.8	51.4	108 02.2	12.9	18 30.6	6.4	54.3	56	16 40	17 20	18 03	13 39	14 17	14 47	15 10	
	15	45 34.9	52.2	122 34.1	12.9	18 24.2	6.4	54.3	58	16 34	17 15	18 02	13 53	14 30	14 58	15 18	
	16	60 35.0	53.0	137 06.0	13.1	18 17.8	6.6	54.3	S 60	16 26	17 11	18 00	14 10	14 45	15 10	15 28	
	17	75 35.1	53.8	151 38.1	13.0	18 11.2	6.6	54.3	DIA SOL ET (+) Pass Pass Merid Idade Fase								
	18	90 35.2	N13 54.6	166 10.1	13.1	S18 04.6	6.7	54.3	00h 12h		Merid		h m	h m	d	%	
	19	105 35.3	55.4	180 42.2	13.2	17 57.9	6.7	54.3	25	01 53	01 59	11 58	04 56	17 21	20 64		
	20	120 35.4	56.2	195 14.4	13.2	17 51.2	6.9	54.3	26	02 04	02 09	11 58	05 46	18 10</			

m 28	SOL PLANETAS	Y	LUA	v ou d Corr.	v ou d Corr.	v ou d Corr.
s	o	i	o	i	i	i
00	7 00.0	7 01.1	6 40.9	0.0 0.0	6.0 2.9	12.0 5.7
01	7 00.3	7 01.4	6 41.1	0.1 0.0	6.1 2.9	12.1 5.7
02	7 00.5	7 01.7	6 41.3	0.2 0.1	6.2 2.9	12.2 5.8
03	7 00.8	7 01.9	6 41.6	0.3 0.1	6.3 3.0	12.3 5.8
04	7 01.0	7 02.2	6 41.8	0.4 0.2	6.4 3.0	12.4 5.9
05	7 01.3	7 02.4	6 42.1	0.5 0.2	6.5 3.1	12.5 5.9
06	7 01.5	7 02.7	6 42.3	0.6 0.3	6.6 3.1	12.6 6.0
07	7 01.8	7 02.9	6 42.5	0.7 0.3	6.7 3.2	12.7 6.0
08	7 02.0	7 03.2	6 42.8	0.8 0.4	6.8 3.2	12.8 6.1
09	7 02.3	7 03.4	6 43.0	0.9 0.4	6.9 3.3	12.9 6.1
10	7 02.5	7 03.7	6 43.3	1.0 0.5	7.0 3.3	13.0 6.2
11	7 02.8	7 03.9	6 43.5	1.1 0.5	7.1 3.4	13.1 6.2
12	7 03.0	7 04.2	6 43.7	1.2 0.6	7.2 3.4	13.2 6.3
13	7 03.3	7 04.4	6 44.0	1.3 0.6	7.3 3.5	13.3 6.3
14	7 03.5	7 04.7	6 44.2	1.4 0.7	7.4 3.5	13.4 6.4
15	7 03.8	7 04.9	6 44.4	1.5 0.7	7.5 3.6	13.5 6.4
16	7 04.0	7 05.2	6 44.7	1.6 0.8	7.6 3.6	13.6 6.5
17	7 04.3	7 05.4	6 44.9	1.7 0.8	7.7 3.7	13.7 6.5
18	7 04.5	7 05.7	6 45.2	1.8 0.9	7.8 3.7	13.8 6.6
19	7 04.8	7 05.9	6 45.4	1.9 0.9	7.9 3.8	13.9 6.6
20	7 05.0	7 06.2	6 45.6	2.0 1.0	8.0 3.8	14.0 6.7
21	7 05.3	7 06.4	6 45.9	2.1 1.0	8.1 3.8	14.1 6.7
22	7 05.5	7 06.7	6 46.1	2.2 1.0	8.2 3.9	14.2 6.7
23	7 05.8	7 06.9	6 46.4	2.3 1.1	8.3 3.9	14.3 6.8
24	7 06.0	7 07.2	6 46.6	2.4 1.1	8.4 4.0	14.4 6.8
25	7 06.3	7 07.4	6 46.8	2.5 1.2	8.5 4.0	14.5 6.9
26	7 06.5	7 07.7	6 47.1	2.6 1.2	8.6 4.1	14.6 6.9
27	7 06.8	7 07.9	6 47.3	2.7 1.3	8.7 4.1	14.7 7.0
28	7 07.0	7 08.2	6 47.5	2.8 1.3	8.8 4.2	14.8 7.0
29	7 07.3	7 08.4	6 47.8	2.9 1.4	8.9 4.2	14.9 7.1
30	7 07.5	7 08.7	6 48.0	3.0 1.4	9.0 4.3	15.0 7.1
31	7 07.8	7 08.9	6 48.3	3.1 1.5	9.1 4.3	15.1 7.2
32	7 08.0	7 09.2	6 48.5	3.2 1.5	9.2 4.4	15.2 7.2
33	7 08.3	7 09.4	6 48.7	3.3 1.6	9.3 4.4	15.3 7.3
34	7 08.5	7 09.7	6 49.0	3.4 1.6	9.4 4.5	15.4 7.3
35	7 08.8	7 09.9	6 49.2	3.5 1.7	9.5 4.5	15.5 7.4
36	7 09.0	7 10.2	6 49.5	3.6 1.7	9.6 4.6	15.6 7.4
37	7 09.3	7 10.4	6 49.7	3.7 1.8	9.7 4.6	15.7 7.5
38	7 09.5	7 10.7	6 49.9	3.8 1.8	9.8 4.7	15.8 7.5
39	7 09.8	7 10.9	6 50.2	3.9 1.9	9.9 4.7	15.9 7.6
40	7 10.0	7 11.2	6 50.4	4.0 1.9	10.0 4.8	16.0 7.6
41	7 10.3	7 11.4	6 50.6	4.1 1.9	10.1 4.8	16.1 7.6
42	7 10.5	7 11.7	6 50.9	4.2 2.0	10.2 4.8	16.2 7.7
43	7 10.8	7 11.9	6 51.1	4.3 2.0	10.3 4.9	16.3 7.7
44	7 11.0	7 12.2	6 51.4	4.4 2.1	10.4 4.9	16.4 7.8
45	7 11.3	7 12.4	6 51.6	4.5 2.1	10.5 5.0	16.5 7.8
46	7 11.5	7 12.7	6 51.8	4.6 2.2	10.6 5.0	16.6 7.9
47	7 11.8	7 12.9	6 52.1	4.7 2.2	10.7 5.1	16.7 7.9
48	7 12.0	7 13.2	6 52.3	4.8 2.3	10.8 5.1	16.8 8.0
49	7 12.3	7 13.4	6 52.6	4.9 2.3	10.9 5.2	16.9 8.0
50	7 12.5	7 13.7	6 52.8	5.0 2.4	11.0 5.2	17.0 8.1
51	7 12.8	7 13.9	6 53.0	5.1 2.4	11.1 5.3	17.1 8.1
52	7 13.0	7 14.2	6 53.3	5.2 2.5	11.2 5.3	17.2 8.2
53	7 13.3	7 14.4	6 53.5	5.3 2.5	11.3 5.4	17.3 8.2
54	7 13.5	7 14.7	6 53.8	5.4 2.6	11.4 5.4	17.4 8.3
55	7 13.8	7 14.9	6 54.0	5.5 2.6	11.5 5.5	17.5 8.3
56	7 14.0	7 15.2	6 54.2	5.6 2.7	11.6 5.5	17.6 8.4
57	7 14.3	7 15.4	6 54.5	5.7 2.7	11.7 5.6	17.7 8.4
58	7 14.5	7 15.7	6 54.7	5.8 2.8	11.8 5.6	17.8 8.5
59	7 14.8	7 15.9	6 54.9	5.9 2.8	11.9 5.7	17.9 8.5
60	7 15.0	7 16.2	6 55.2	6.0 2.9	12.0 5.7	18.0 8.6

m 29	SOL PLANETAS	Y	LUA	v ou d Corr.	v ou d Corr.	v ou d Corr.
s	o	i	o	i	i	i
00	7 15.0	7 16.2	6 55.2	0.0 0.0	6.0 3.0	12.0 5.9
01	7 15.3	7 16.4	6 55.4	0.1 0.0	6.1 3.0	12.1 5.9
02	7 15.5	7 16.7	6 55.7	0.2 0.1	6.2 3.0	12.2 6.0
03	7 15.8	7 16.9	6 55.9	0.3 0.1	6.3 3.1	12.3 6.0
04	7 16.0	7 17.2	6 56.1	0.4 0.2	6.4 3.1	12.4 6.1
05	7 16.3	7 17.4	6 56.4	0.5 0.2	6.5 3.2	12.5 6.1
06	7 16.5	7 17.7	6 56.6	0.6 0.3	6.6 3.2	12.6 6.2
07	7 16.8	7 17.9	6 56.9	0.7 0.3	6.7 3.3	12.7 6.2
08	7 17.0	7 18.2	6 57.1	0.8 0.4	6.8 3.3	12.8 6.3
09	7 17.3	7 18.4	6 57.3	0.9 0.4	6.9 3.4	12.9 6.3
10	7 17.5	7 18.7	6 57.6	1.0 0.5	7.0 3.4	13.0 6.4
11	7 17.8	7 18.9	6 57.8	1.1 0.5	7.1 3.5	13.1 6.4
12	7 18.0	7 19.2	6 58.0	1.2 0.6	7.2 3.5	13.2 6.5
13	7 18.3	7 19.4	6 58.3	1.3 0.6	7.3 3.6	13.3 6.5
14	7 18.5	7 19.7	6 58.5	1.4 0.7	7.4 3.6	13.4 6.6
15	7 18.8	7 20.0	6 58.8	1.5 0.7	7.5 3.7	13.5 6.6
16	7 19.0	7 20.2	6 59.0	1.6 0.8	7.6 3.7	13.6 6.7
17	7 19.3	7 20.5	6 59.2	1.7 0.8	7.7 3.8	13.7 6.7
18	7 19.5	7 20.7	6 59.5	1.8 0.9	7.8 3.8	13.8 6.8
19	7 19.8	7 21.0	6 59.7	1.9 0.9	7.9 3.9	13.9 6.8
20	7 20.0	7 21.2	7 00.0	2.0 1.0	8.0 3.9	14.0 6.9
21	7 20.3	7 21.5	7 00.2	2.1 1.0	8.1 4.0	14.1 6.9
22	7 20.5	7 21.7	7 00.4	2.2 1.1	8.2 4.0	14.2 7.0
23	7 20.8	7 22.0	7 00.7	2.3 1.1	8.3 4.1	14.3 7.0
24	7 21.0	7 22.2	7 00.9	2.4 1.2	8.4 4.1	14.4 7.1
25	7 21.3	7 22.5	7 01.1	2.5 1.2	8.5 4.2	14.5 7.1
26	7 21.5	7 22.7	7 01.4	2.6 1.3	8.6 4.2	14.6 7.2
27	7 21.8	7 23.0	7 01.6	2.7 1.3	8.7 4.3	14.7 7.2
28	7 22.0	7 23.2	7 01.9	2.8 1.4	8.8 4.3	14.8 7.3
29	7 22.3	7 23.5	7 02.1	2.9 1.4	8.9 4.4	14.9 7.3
30	7 22.5	7 23.7	7 02.3	3.0 1.5	9.0 4.4	15.0 7.4
31	7 22.8	7 24.0	7 02.6	3.1 1.5	9.1 4.5	15.1 7.4
32	7 23.0	7 24.2	7 02.8	3.2 1.6	9.2 4.5	15.2 7.5
33	7 23.3	7 24.5	7 03.1	3.3 1.6	9.3 4.6	15.3 7.5
34	7 23.5	7 24.7	7 03.3	3.4 1.7	9.4 4.6	15.4 7.6
35	7 23.8	7 25.0	7 03.5	3.5 1.7	9.5 4.7	15.5 7.6
36	7 24.0	7 25.2	7 03.8	3.6 1.8	9.6 4.7	15.6 7.7
37	7 24.3	7 25.5	7 04.0	3.7 1.8	9.7 4.8	15.7 7.7
38	7 24.5	7 25.7	7 04.3	3.8 1.9	9.8 4.8	15.8 7.8
39	7 24.8	7 26.0	7 04.5	3.9 1.9	9.9 4.9	15.9 7.8
40	7 25.0	7 26.2	7 04.7	4.0 2.0	10.0 4.9	16.0 7.9
41	7 25.3	7 26.5	7 05.0	4.1 2.0	10.1 5.0	16.1 7.9
42	7 25.5	7 26.7	7 05.2	4.2 2.1	10.2 5.0	16.2 8.0
43	7 25.8	7 27.0	7 05.4	4.3 2.1	10.3 5.1	16.3 8.0
44	7 26.0	7 27.2	7 05.7	4.4 2.2	10.4 5.1	16.4 8.1
45	7 26.3	7 27.5	7 05.9	4.5 2.2	10.5 5.2	16.5 8.1
46	7 26.5	7 27.7	7 06.2	4.6 2.3	10.6 5.2	16.6 8.2
47	7 26.8	7 28.0	7 06.4	4.7 2.3	10.7 5.3	16.7 8.2
48	7 27.0	7 28.2	7 06.6	4.8 2.4	10.8 5.3	16.8 8.3
49	7 27.3	7 28.5	7 06.9	4.9 2.4	10.9 5.4	16.9 8.3
50	7 27.5	7 28.7	7 07.1	5.0 2.5	11.0 5.4	17.0 8.4
51	7 27.8	7 29.0	7 07.4	5.1 2.5	11.1 5.5	17.1 8.4
52	7 28.0	7 29.2	7 07.6	5.2 2.6	11.2 5.5	17.2 8.5
53	7 28.3	7 29.5	7 07.8	5.3 2.6	11.3 5.6	17.3 8.5
54	7 28.5	7 29.7	7 08.1	5.4 2.7	11.4 5.6	17.4 8.6
55	7 28.8	7 30.0	7 08.3	5.5 2.7	11.5 5.7	17.5 8.6
56	7 29.0	7 30.2	7 08.5	5.6 2.8	11.6 5.7	17.6 8.7
57	7 29.3	7 30.5	7 08.8	5.7 2.8	11.7 5.8	17.7 8.7
58	7 29.5	7 30.7	7 09.0	5.8 2.9	11.8 5.8	17.8 8.8
59	7 29.8	7 31.0	7 09.3	5.9 2.9	11.9 5.9	17.9 8.8
60	7 30.0	7 31.2	7 09.5	6.0 3.0	12.0 5.9	18.0 8.9

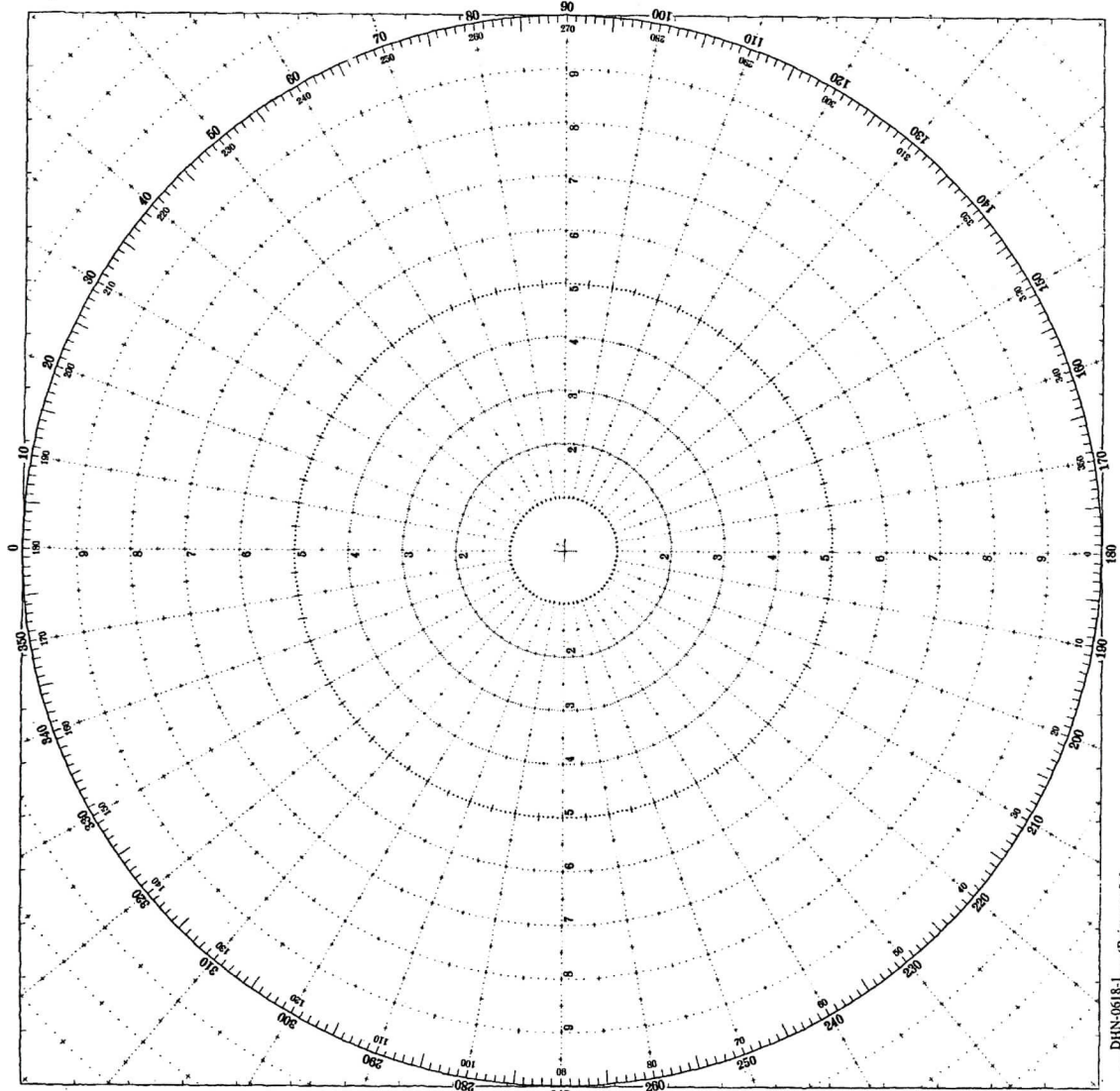
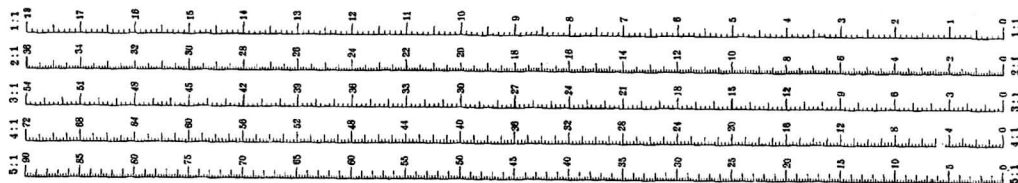
CONVERSÃO DE ARCO EM TEMPO

0° - 59°		60° - 119°		120° - 179°		180° - 239°		240° - 299°		300° - 359°			0'00	0'25	0'50	0'75
°	h m	°	h m	°	h m	°	h m	°	h m	°	h m	°	m s	m s	m s	m s
0	0 00	60	4 00	120	8 00	180	12 00	240	16 00	300	20 00	0	0 00	0 01	0 02	0 03
1	0 04	61	4 04	121	8 04	181	12 04	241	16 04	301	20 04	1	0 04	0 05	0 06	0 07
2	0 08	62	4 08	122	8 08	182	12 08	242	16 08	302	20 08	2	0 08	0 09	0 10	0 11
3	0 12	63	4 12	123	8 12	183	12 12	243	16 12	303	20 12	3	0 12	0 13	0 14	0 15
4	0 16	64	4 16	124	8 16	184	12 16	244	16 16	304	20 16	4	0 16	0 17	0 18	0 19
5	0 20	65	4 20	125	8 20	185	12 20	245	16 20	305	20 20	5	0 20	0 21	0 22	0 23
6	0 24	66	4 24	126	8 24	186	12 24	246	16 24	306	20 24	6	0 24	0 25	0 26	0 27
7	0 28	67	4 28	127	8 28	187	12 28	247	16 28	307	20 28	7	0 28	0 29	0 30	0 31
8	0 32	68	4 32	128	8 32	188	12 32	248	16 32	308	20 32	8	0 32	0 33	0 34	0 35
9	0 36	69	4 36	129	8 36	189	12 36	249	16 36	309	20 36	9	0 36	0 37	0 38	0 39
10	0 40	70	4 40	130	8 40	190	12 40	250	16 40	310	20 40	10	0 40	0 41	0 42	0 43
11	0 44	71	4 44	131	8 44	191	12 44	251	16 44	311	20 44	11	0 44	0 45	0 46	0 47
12	0 48	72	4 48	132	8 48	192	12 48	252	16 48	312	20 48	12	0 48	0 49	0 50	0 51
13	0 52	73	4 52	133	8 52	193	12 52	253	16 52	313	20 52	13	0 52	0 53	0 54	0 55
14	0 56	74	4 56	134	8 56	194	12 56	254	16 56	314	20 56	14	0 56	0 57	0 58	0 59
15	1 00	75	5 00	135	9 00	195	13 00	255	17 00	315	21 00	15	1 00	1 01	1 02	1 03
16	1 04	76	5 04	136	9 04	196	13 04	256	17 04	316	21 04	16	1 04	1 05	1 06	1 07
17	1 08	77	5 08	137	9 08	197	13 08	257	17 08	317	21 08	17	1 08	1 09	1 10	1 11
18	1 12	78	5 12	138	9 12	198	13 12	258	17 12	318	21 12	18	1 12	1 13	1 14	1 15
19	1 16	79	5 16	139	9 16	199	13 16	259	17 16	319	21 16	19	1 16	1 17	1 18	1 19
20	1 20	80	5 20	140	9 20	200	13 20	260	17 20	320	21 20	20	1 20	1 21	1 22	1 23
21	1 24	81	5 24	141	9 24	201	13 24	261	17 24	321	21 24	21	1 24	1 25	1 26	1 27
22	1 28	82	5 28	142	9 28	202	13 28	262	17 28	322	21 28	22	1 28	1 29	1 30	1 31
23	1 32	83	5 32	143	9 32	203	13 32	263	17 32	323	21 32	23	1 32	1 33	1 34	1 35
24	1 36	84	5 36	144	9 36	204	13 36	264	17 36	324	21 36	24	1 36	1 37	1 38	1 39
25	1 40	85	5 40	145	9 40	205	13 40	265	17 40	325	21 40	25	1 40	1 41	1 42	1 43
26	1 44	86	5 44	146	9 44	206	13 44	266	17 44	326	21 44	26	1 44	1 45	1 46	1 47
27	1 48	87	5 48	147	9 48	207	13 48	267	17 48	327	21 48	27	1 48	1 49	1 50	1 51
28	1 52	88	5 52	148	9 52	208	13 52	268	17 52	328	21 52	28	1 52	1 53	1 54	1 55
29	1 56	89	5 56	149	9 56	209	13 56	269	17 56	329	21 56	29	1 56	1 57	1 58	1 59
30	2 00	90	6 00	150	10 00	210	14 00	270	18 00	330	22 00	30	2 00	2 01	2 02	2 03
31	2 04	91	6 04	151	10 04	211	14 04	271	18 04	331	22 04	31	2 04	2 05	2 06	2 07
32	2 08	92	6 08	152	10 08	212	14 08	272	18 08	332	22 08	32	2 08	2 09	2 10	2 11
33	2 12	93	6 12	153	10 12	213	14 12	273	18 12	333	22 12	33	2 12	2 13	2 14	2 15
34	2 16	94	6 16	154	10 16	214	14 16	274	18 16	334	22 16	34	2 16	2 17	2 18	2 19
35	2 20	95	6 20	155	10 20	215	14 20	275	18 20	335	22 20	35	2 20	2 21	2 22	2 23
36	2 24	96	6 24	156	10 24	216	14 24	276	18 24	336	22 24	36	2 24	2 25	2 26	2 27
37	2 28	97	6 28	157	10 28	217	14 28	277	18 28	337	22 28	37	2 28	2 29	2 30	2 31
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39	2 36	99	6 36	159	10 36	219	14 36	279	18 36	339	22 36	39	2 36	2 37	2 38	2 39
40	2 40	100	6 40	160	10 40	220	14 40	280	18 40	340	22 40	40	2 40	2 41	2 42	2 43
41	2 44	101	6 44	161	10 44	221	14 44	281	18 44	341	22 44	41	2 44	2 45	2 46	2 47
42	2 48	102	6 48	162	10 48	222	14 48	282	18 48	342	22 48	42	2 48	2 49	2 50	2 51
43	2 52	103	6 52	163	10 52	223	14 52	283	18 52	343	22 52	43	2 52	2 53	2 54	2 55
44	2 56	104	6 56	164	10 56	224	14 56	284	18 56	344	22 56	44	2 56	2 57	2 58	2 59
45	3 00	105	7 00	165	11 00	225	15 00	285	19 00	345	23 00	45	3 00	3 01	3 02	3 03
46	3 04	106	7 04	166	11 04	226	15 04	286	19 04	346	23 04	46	3 04	3 05	3 06	3 07
47	3 08	107	7 08	167	11 08	227	15 08	287	19 08	347	23 08	47	3 08	3 09	3 10	3 11
48	3 12	108	7 12	168	11 12	228	15 12	288	19 12	348	23 12	48	3 12	3 13	3 14	3 15
49	3 16	109	7 16	169	11 16	229	15 16	289	19 16	349	23 16	49	3 16	3 17	3 18	3 19
50	3 20	110	7 20	170	11 20	230	15 20	290	19 20	350	23 20	50	3 20	3 21	3 22	3 23
51	3 24	111	7 24	171	11 24	231	15 24	291	19 24	351	23 24	51	3 24	3 25	3 26	3 27
52	3 28	112	7 28	172	11 28	232	15 28	292	19 28	352	23 28	52	3 28	3 29	3 30	3 31
53	3 32	113	7 32	173	11 32	233	15 32	293	19 32	353	23 32	53	3 32	3 33	3 34	3 35
54	3 36	114	7 36	174	11 36	234	15 36	294	19 36	354	23 36	54	3 36	3 37	3 38	3 39
55	3 40	115	7 40	175	11 40	235	15 40	295	19 40	355	23 40	55	3 40	3 41	3 42	3 43
56	3 44	116	7 44	176	11 44	236	15 44	296	19 44	356	23 44	56	3 44	3 45	3 46	3 47
57	3 48	117	7 48	177	11 48	237	15 48	297	19 48	357	23 48	57	3 48	3 49	3 50	3 51
58	3 52	118	7 52	178	11 52	238	15 52	298	19 52	358	23 52	58	3 52	3 53	3 54	3 55
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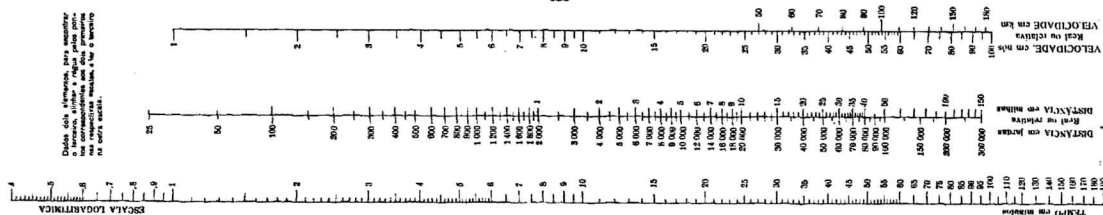
A tabela acima destina-se à conversão de arco em tempo; sua principal aplicação nesse Almanaque é a conversão da longitude, cujo valor em horas, minutos e segundos é utilizado na fórmula que relaciona a HML com a TU: $TU = HML + \lambda$, sendo λ positivo para longitude W e negativo para longitude E.



ESCALAS



DHN-0618.1 (Reimpresso)
15.000-VI-2005



Para a leitura correta, para qualquer
distância, velocidade ou tempo, deve-se
usar a mesma unidade de medida em
todas as escalas.